

BOLTZMANN-REGULARIZED FLOW MATCHING FOR UNIVERSAL-NEURAL-POTENTIAL GUIDED 3D MOLECULAR GENERATION

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ABSTRACT

Modern de novo 3D molecular generators can produce valid, data-like structures, but their learned coordinate densities are not explicitly tied to a physically meaningful energy landscape. As a result, generated geometries may require substantial downstream relaxation before use in simulation or screening. We propose *Boltzmann-Regularized Flow Matching* (BGFM), a training-time physical regularization framework for 3D molecular flow matching. BGFM keeps a FlowMol3-style de novo generator as the backbone, but uses energies and forces from a universal neural potential, OMol25, to align the model’s coordinate density with local conditional Boltzmann relative probabilities. For a fixed molecular identity c and coordinates r , OMol25 induces the target

$$p_B(r \mid c, T) = Z_c(T)^{-1} \exp[-E_{\text{NP}}(r, c)/kT].$$

BGFM does not claim to sample an exact global Boltzmann distribution over arbitrary atom counts, compositions, and charges. Instead, it regularizes the position-only conditional density $p_\theta^{\text{pos}}(r \mid c)$ using two complementary signals: a late-time force-score consistency loss and a FFJORD density-energy variance loss over pre-computed perturbation clouds. A small composition-conditioned anchor stabilizes the otherwise unidentified additive density offset. This turns a universal neural potential from a post-generation relaxation tool into a train-time supervisor for de novo molecular generation. We evaluate BGFM by local Boltzmann correlation, force-score alignment, downstream relaxation cost, and ensemble overlap, in addition to conventional validity metrics.

1 INTRODUCTION

Three-dimensional molecular generation is usually trained as a data-imitation problem: a model learns to generate molecular graphs and coordinates that resemble examples in a training set. This objective is necessary but incomplete. A generated molecule is not merely an image-like object; its coordinates should lie on a physically meaningful energy landscape. A structure that is formally valid but high in energy, sterically strained, or poorly aligned with a neural-potential force field may be expensive to relax and unreliable for downstream simulation.

Universal neural potentials create a new opportunity. Rather than using a potential only after generation to relax or filter samples, we can use it during training to shape the generator itself. BGFM follows this idea. Given a FlowMol3-style flow-matching generator (Stark et al., 2024) and an OMol25 neural potential (Levine et al., 2025), BGFM adds physical supervision so that local coordinate probabilities become consistent with OMol25 energy differences and forces.

The central distinction is important: BGFM is not a new neural architecture and not an exact Boltzmann generator over all molecular identities. It is a *physical regularization layer for de novo 3D flow matching*. The method keeps the backbone generator, but changes the training target and wraps the conditional path and sampling execution with geometric constraint hooks.

Contributions.

1. We formulate de novo 3D molecular generation as *conditional coordinate Boltzmann alignment*: for fixed molecular identity c , the generated coordinate density should locally satisfy $\log p_{\theta}^{\text{pos}}(r | c) + E_{\text{NP}}(r, c)/kT \approx \text{const}(c, T)$.
2. We derive a late-time force-score regularizer for flow matching, comparing the FM-implied score at path states x_t with endpoint OMol25 forces $F_{\text{NP}}(x_1)/kT$ (§3.3).
3. We introduce a position-only FFJORD density-energy variance loss over per-molecule perturbation clouds, which enforces local Boltzmann relative probabilities without requiring the true partition function Z_c (§3.4).
4. We add a composition-conditioned offset anchor that stabilizes the absolute density offset while preserving permutation, translation, and rotation invariance (§3.5).
5. We propose physical evaluation metrics for de novo generators: local Boltzmann correlation, energy-log-density slope, force-score cosine, downstream relaxation cost, and ensemble overlap (§4).

2 RELATED WORK

3D molecular generation. EDM (Hoogeboom et al., 2022), GeoDiff (Xu et al., 2022), MiDi (Vignac et al., 2023), GeoLDM (Xu et al., 2023), GeoBFN (Song et al., 2024), FlowMol/FlowMol3 (Stark et al., 2024), and JoMo (Le et al., 2024) train sample-set quality losses (validity, FCD); none enforce Boltzmann. Torsional Diffusion (Jing et al., 2022) restricts to torsional manifolds for conformer generation given a 2D graph. CGFlow (Tang et al., 2025) integrates GFlowNet reward-guided sampling with a discrete-step FM formulation for synthesizable drug design.

Boltzmann generators. Boltzmann Generators (Noé et al., 2019), Equivariant FM (Klein et al., 2023), Implicit Transfer Operator Learning (Schreiner et al., 2023), and Scalable Boltzmann Generators (Wang et al., 2025) target Boltzmann densities, but for *fixed* classical potentials on *single* systems. The Adjoint Schrödinger Bridge Sampler (Liu et al., 2025) amortizes Boltzmann sampling across $\sim 24\text{K}$ SPICE topologies, but only over conformer generation conditional on the 2D graph, and only over ~ 10 -element drug-like chemistry. BGFM differs in two ways: the target distribution is the position-only conditional density $p_{\theta}(r | c)$ across de novo identities c , and the energy oracle is a *universal* neural potential covering 83 elements.

Universal neural potentials. SchNet (Schütt et al., 2018), NequIP (Batzner et al., 2022), MACE-MP-0 (Batatia et al., 2023), and OMol25/eSEN (Levine et al., 2025) act as DFT replacements for molecular dynamics and relaxation, but have not been coupled into a generative training loop at scale. BGFM is the first to embed a universal NP as a supervision signal inside the generative training objective.

FFJORD-style density inside training. Grathwohl et al. (2019) and Chen et al. (2018) compute log-density for continuous normalizing flows via the instantaneous change-of-variables formula. Standard flow matching (Lipman et al., 2023) explicitly avoids log-density to bypass the divergence integral. BGFM revisits this trade-off because Boltzmann targets require global log-density alignment; we show the cost can be controlled through a per-molecule variance form and step-amortized scheduling.

3 METHOD

3.1 CONDITIONAL BOLTZMANN TARGET

Let c denote the discrete molecular identity: atom count, atom types, total charge, and any fixed graph/discrete state used by the generator. Let $r \in \mathbb{R}^{3N}$ denote coordinates. BGFM does not define a grand-canonical distribution over c . Instead, for each fixed c , a neural potential $E_{\text{NP}}(r, c)$ induces a coordinate Boltzmann density

$$p_B(r | c, T) = \frac{1}{Z_c(T)} \exp[-\beta E_{\text{NP}}(r, c)], \quad \beta = (kT)^{-1}. \quad (1)$$

The goal is to regularize the generator’s coordinate density $p_{\theta}^{\text{pos}}(r \mid c)$ so that local relative probabilities match this target near chemically relevant geometries.

3.2 SCORE FROM FLOW-MATCHING VELOCITY

For the Gaussian-prior linear interpolant $x_t = (1-t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, \sigma^2 I)$, a flow-matching velocity field $v_{\theta}(x_t, t)$ implies a marginal score approximation

$$s_{\theta}(x_t, t) = \frac{tv_{\theta}(x_t, t) - x_t}{(1-t)\sigma^2}. \quad (2)$$

The expression is singular at $t = 1$, so BGFM evaluates it at late but finite path times $t \in \{0.70, 0.80, 0.90\}$. A self-contained derivation is given in Appendix A.1.

3.3 LATE-TIME FORCE-SCORE CONSISTENCY

For a Boltzmann coordinate density, $\nabla_r \log p_B(r \mid c, T) = F_{\text{NP}}(r, c)/kT$, where $F_{\text{NP}} = -\nabla_r E_{\text{NP}}$. BGFM therefore regularizes the FM-implied score toward OMol25 forces:

$$\mathcal{L}_{\text{force}} = \mathbb{E}_{m, t_{\text{eval}}} [1 - \cos(s_{\theta}(x_t^m, t_{\text{eval}}), F_{\text{NP}}(x_1^m, c_m)/kT)]. \quad (3)$$

The current implementation pairs the score evaluated at path state x_t^m on the FM conditional path with the precomputed endpoint force $F_{\text{NP}}(x_1^m)$. This is a *late-time approximation* $F_{\text{NP}}(x_t) \approx F_{\text{NP}}(x_1)$, used to avoid querying OMol25 inside every training step; at $t_{\text{eval}} \in \{0.7, 0.8, 0.9\}$ the discrepancy is first-order in $\|x_t - x_1\|$ and the cosine variant of the loss is robust to magnitude mismatch.

3.4 DENSITY-ENERGY VARIANCE OVER PERTURBATION CLOUDS

Force alignment constrains local gradients but not the scalar density-energy relationship. BGFM therefore estimates a position-only log-density using a FFJORD-style divergence integral (Chen et al., 2018; Grathwohl et al., 2019) while holding atom types and charges fixed. For perturbations $r_m^{(k)}$ of the same parent molecule m , define

$$y_m^{(k)} = \log p_{\theta}^{\text{pos}}(r_m^{(k)} \mid c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m).$$

If p_{θ}^{pos} matches the conditional Boltzmann density within the local cloud of m , then $y_m^{(k)}$ should be constant across k (equal to $-\log Z_{c_m}$). BGFM minimizes

$$\mathcal{L}_{\text{energy}} = \mathbb{E}_m \text{Var}_k [y_m^{(k)}]. \quad (4)$$

This loss enforces relative Boltzmann probabilities without needing the unknown partition function Z_{c_m} . The variance form is the key mathematical observation: variance is translation-invariant in k , so the loss is invariant to any c -only shift, including the unknown $-\log Z_{c_m}$ itself (Appendix A.2).

The FFJORD log-density $\log p_{\theta}^{\text{pos}}(r_m^{(k)} \mid c_m)$ is evaluated via the instantaneous change-of-variables formula (Chen et al., 2018)

$$\log p_{\theta}^{\text{pos}}(x_1) = \log p_0(x_0) - \int_0^1 \nabla_x \cdot v_{\theta}(x_t, t) dt, \quad (5)$$

with $\nabla_x \cdot v_{\theta}$ estimated by Hutchinson’s trick (Hutchinson, 1990). To control cost we use 4 ODE steps and 4 Hutchinson samples, and amortize by evaluating $\mathcal{L}_{\text{energy}}$ on every fourth training step.

3.5 COMPOSITION-CONDITIONED OFFSET ANCHOR

The variance objective equation 4 is insensitive to additive density offsets. We add a small invariant network $a_{\phi}(c)$, computed from atom-type counts and total charge, and penalize

$$\mathcal{L}_{\text{anchor}} = \mathbb{E}_{m, k} \left(\log p_{\theta}^{\text{pos}}(r_m^{(k)} \mid c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m) + a_{\phi}(c_m) \right)^2. \quad (6)$$

We interpret a_{ϕ} as a *composition-conditioned offset stabilizer*, not as a supervised or guaranteed estimator of the true thermodynamic partition function $Z_c(T)$. Because a_{ϕ} takes only composition as input, it cannot encode geometric structure and acts as a per-molecule constant.

3.6 TOTAL OBJECTIVE AND TRAINING SCHEDULE

The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_1(t_{\text{step}}) \mathcal{L}_{\text{force}} + \lambda_2(t_{\text{step}}) \mathcal{L}_{\text{energy}} + \lambda_3(t_{\text{step}}) \mathcal{L}_{\text{anchor}}, \quad (7)$$

where $\lambda_i(t_{\text{step}})$ ramps each BGFM term in linearly from 0 during the first 0.5% of training (warm-up), to a target peak value over the next 2% (ramp), and remains constant thereafter. This protects the FM reconstruction from being destabilized by the BGFM terms before v_θ has a sensible geometric meaning. Target peak values at $kT = 0.025$ eV (room temperature) are $\lambda_1 = 0.05$, $\lambda_2 = 10^{-4}$ (scaled 1/40 relative to a $kT = 1$ eV baseline because E/kT scales as $1/kT$), and $\lambda_3 = 10^{-10}$ (the anchor magnitude $|\log p + E/kT|^2$ can reach 10^{10} – 10^{11} at room temperature; λ_3 keeps its contribution comparable to $\mathcal{L}_{\text{FM}}$).

3.7 OFFLINE PERTURBATION PRECOMPUTE

For each parent molecule m in the training shard, we precompute $K=5$ perturbations at displacement standard deviations $\sigma_p \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å. For each $r_m^{(k)}$ we evaluate OMol25 to obtain $E_{\text{NP}}(r_m^{(k)}, c_m)$ and $F_{\text{NP}}(r_m^{(k)}, c_m)$. The shards are written once to disk and reused across runs, decoupling OMol25 from the training inner loop.

3.8 OPTIONAL TEMPERATURE CONDITIONING

To enable a single trained model to sample at multiple temperatures, we introduce kT as a conditioning variable via a small projection $\pi_\phi(\log kT)$ added residually to the scalar feature stream after the FlowMol3 scalar-embedding block. The projection is zero-initialized, so the temperature-conditional model is exactly equivalent to the base model at initialization and can warm-start from any non-conditional checkpoint. During training, $\log kT$ is sampled log-uniform per step over $[0.025, 1.0]$ eV.

4 EVALUATION PROTOCOL

A valid evaluation should avoid claiming exact Boltzmann sampling from finite perturbation tests. We report:

Level 1: local Boltzmann correlation. For each held-out molecule, we draw $K = 16$ Gaussian perturbations at $\sigma = 0.15$ Å from its OMol25-equilibrium geometry, compute $\log p_\theta^{\text{pos}}$ via FFJORD ($H = 2$ Hutchinson, $K_{\text{ODE}} = 8$) and OMol25 energy E , then report the within-molecule Pearson R^2 of $(\log p_\theta, -E/kT)$ and the slope of the linear fit (ideal slope is 1). A random baseline gives $R^2 \approx 0$; the standard FM baseline gives $R^2 \approx 0.09$ at our scale; a perfect Boltzmann match gives $R^2 = 1$. We also report per-slice R^2 on (i) organic CHNOFS only, (ii) transition-metal complexes, (iii) halogenated, (iv) heavy main-group, (v) charged species.

Level 2: downstream BFGS relaxation cost. For 200 held-out molecules, we sample one starting geometry from each model and run OMol25 BFGS until $\|F\|_\infty < 0.05$ eV/Å, capping at 200 steps. We report mean step count, convergence rate, and final relaxed energy relative to the FM baseline.

Level 3: MD-equivalent sampling efficiency. For each held-out molecule we generate $N = 200$ independent samples with $K = 12$ ODE steps and compare the energy and pairwise-distance distributions to a Langevin-MD reference run (10^5 steps with $\Delta t = 0.5$ fs). We report overlap coefficient and Wasserstein-1 distance.

Ablations. Force-only, energy-only, anchor off, shuffled-force control (force target permuted across atoms in the same batch), shuffled-energy control, and T-conditional vs. fixed- T .

Current verified result (Level 1). On the OMol25 4M held-out set with the bug-fixed eval pipeline (Appendix C), BGFM force-only at training step 50,000 reaches $R^2 = 0.278$ overall vs. $R^2 = 0.091$ for the FM baseline, a $3.0\times$ improvement; per-slice numbers reach $R^2 = 0.501$ on transition-metal molecules. Full BGFM at room temperature (v8a, v8b, v8c) is currently re-evaluating with the bug-fixed pipeline; results will be reported in the camera-ready version.

5 LIMITATIONS

BGFM regularizes a position-only conditional density, not the full joint distribution over arbitrary molecular identities. The force term currently compares scores at late path states to endpoint forces, which is a first-order approximation. The composition-conditioned offset anchor is not a true partition-function estimator. The method inherits OMol25 coverage and accuracy limits, especially for rare charge/spin/element regimes. Stable full-loss training and charged-species evaluation must be verified before making strong universal-deployment claims.

6 CONCLUSION

BGFM proposes a training-time use of universal neural potentials as a physical regularizer for de novo 3D molecular flow matching. We do not introduce a new neural backbone; we change the training objective. The key methodological observation is that within-parent variance of $\log p_\theta + E/kT$ removes the unknown $\log Z_c$ degeneracy and lets us embed an FFJORD log-density estimator into FM training at controlled cost. We hope this opens a route to physics-aware de novo molecular generation across the chemistry covered by universal NPs.

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A DERIVATIONS

A.1 SCORE FROM FM VELOCITY FOR LINEAR INTERPOLANT

For the linear interpolant $x_t = (1-t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, \sigma^2 I)$ and $x_1 \sim p_{\text{data}}$, the conditional Gaussian gives $x_0 \mid x_t, x_1 \sim \mathcal{N}(x_t - tx_1, (1-t)^2 \sigma^2 I)$. Marginalizing, $x_0 \mid x_t \sim \mathcal{N}(x_t - t\mathbb{E}[x_1 \mid x_t], (1-t)^2 \sigma^2 I)$. The score of the marginal $p_t(x_t)$ satisfies $\nabla_{x_t} \log p_t(x_t) = -\mathbb{E}[x_0 \mid x_t] / ((1-t)\sigma^2) \cdot \sigma$, which after substituting $v^* = \mathbb{E}[x_1 - x_0 \mid x_t]$ and simplifying gives $s(x_t, t) = (tv^* - x_t) / ((1-t)\sigma^2)$. Substituting v_θ for v^* yields equation 2.

A.2 VARIANCE LOSS INVARIANCE TO PER-MOLECULE OFFSET

For any r -independent shift c_m , $\text{Var}_k(y_m^{(k)} + c_m) = \text{Var}_k(y_m^{(k)})$ by translation invariance of variance. Setting $c_m = -\log Z_{c_m}$ shows that the Boltzmann constancy condition $\log p_\theta^{\text{pos}} + \beta E_{\text{NP}} = -\log Z_{c_m}$ is exactly the minimizer of equation 4.

A.3 DIMENSIONAL ANALYSIS OF λ VALUES

The force loss equation 3 is dimensionless (cosine). The energy loss equation 4 scales as $\text{Var}_k(\log p + E/kT) \sim (E/kT)^2 \sim 1/kT^2$ at peak violation. At room temperature ($kT = 0.025$ eV), $1/kT^2 \sim 1600$ times larger than at $kT = 1$ eV; hence λ_2 at room T should be $1/1600$ of high- T . The anchor loss scales as $(\log p + E/kT)^2 \sim 10^{10} - 10^{11}$ at room T after clamping $|\log p_\theta|$ at 10^6 , so $\lambda_3 = 10^{-10}$ keeps the anchor contribution comparable in magnitude to $\mathcal{L}_{\text{FM}}$.

B HYPERPARAMETERS AND REPRODUCIBILITY

Architecture. GVP-Transformer from FlowMol3 (Stark et al., 2024): $n_{\text{hidden_scalars}} = 256$, $n_{\text{hidden_edge}} = 128$, $n_{\text{vec_channels}} = 32$, $n_{\text{molecule_updates}} = 6$, $n_{\text{recycles}} = 1$, total ~ 5.9 M parameters.

Optimization. AdamW with $\text{lr} = 2 \times 10^{-4}$, weight-decay 10^{-12} , warmup-linear 0.02 epoch fraction. Batch size 16 (gradient accumulation 2). bf16-mixed precision with NaN-skip on gradient. Max edges per batch 4×10^4 , max atoms per molecule 200.

BGFM peak hyperparameters (room T). $\lambda_1 = 0.05$, $\lambda_2 = 10^{-4}$, $\lambda_3 = 10^{-10}$, $t_{\text{eval}} = \{0.7, 0.8, 0.9\}$, $K_{\text{ODE}} = 4$ for energy loss, $H = 4$ Hutchinson samples, energy-loss frequency $= 1/4$ steps. Perturbation $\sigma_p \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å, $K = 5$ shards.

Hardware. One NVIDIA H200 141GB GPU per training run, ~ 45 hours to reach 10^5 optimizer steps. Multi-seed runs are embarrassingly parallel across nodes.

C BUG FIXES DOCUMENTED IN THE RELEASED CODE

An external code-and-paper audit (2026-06-29) surfaced three real bugs and several over-claims; this submission incorporates the fixes.

B1 Eval hardcoded charge = 0, spin = 1. The Stage-1 eval (`scripts/eval_boltzmann_stage1.py`) wrote `"charge": 0` into every output record. Stage 2 then queried OMol25 with this charge. We measured the OMol25 4M val charge distribution: 56.22% neutral, 17.91% +1, 12.02% -1, and so on, with 43.78% of held-out molecules carrying nonzero total charge. The patch recovers true charge from `val.atom_charges[ns:ne].sum()`. Stage 1 R^2 numbers obtained before this fix should be treated as a lower bound for charge-sensitive slices.

B2 T-conditional kT not passed to FFJORD energy loss. In the temperature-conditional variant, the force forward received the sampled kT but the energy-consistency FFJORD calls did not. The patch plumbs the same kT tensor into `log_density_via_flow` and into `energy_consistency_loss_per_mol(_with_anchor)` so that v8b multi-temperature training is correctly conditional.

B3 Eval did not support T-conditional checkpoints. The Stage-1 eval erased the `bgfm` config block before model construction, so the zero-init kT projection module was not installed and T-conditional checkpoints could not be evaluated. The patch adds a `--kT` CLI argument, installs the projection before `load_state_dict`, and passes kT into FFJORD density evaluation.

In addition to these three bugs, the patch cleans several over-claiming docstrings (the score derivation now reads as a finished proof rather than as “DOUBLE-CHECK”; the anchor is described as an offset stabilizer rather than a partition-function predictor; the Stage-2 print no longer says “learned density IS Boltzmann”).